

Title

Facility-Level Variation and Barriers to Guideline-Adherent Rehabilitation Intensity

Abstract

We examined facility-level variation in rehabilitation intensity across eight participating Canadian centers. Mean aerobic exercise time varied from 6 minutes to 18 minutes per session across sites (range 40-64% of participants achieving guideline-recommended minimum intensity levels). Facilities with lower mean aerobic minutes tended to have larger patient caseloads per therapist and shorter session durations. Session structure analysis revealed that time spent on functional assessment, rest breaks, and patient transportation accounted for 30-45% of scheduled rehabilitation time, leaving 55-70% for active therapy. Participants with greater baseline functional capacity and those receiving extended session durations (≥ 60 minutes) achieved higher aerobic minutes. Qualitative observations suggested perceived time constraints and reimbursement limitations as reported barriers to higher-intensity protocols. Facilities that provided real-time biofeedback to therapists (via displays of heart rate and step data during sessions) showed non-significantly higher mean aerobic minutes (13 vs 10 minutes, $p = 0.11$). These facility and structural factors may represent modifiable barriers to implementing guideline-recommended rehabilitation intensity.